

Random Fields Supporting Quasi-Zero Modes in Nonperturbative QED: Connection with Asymptotic Freedom

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It is found that any QED process calculated from a functional integral requires new, non-perturbative terms in its effective action. These are obtained from a rearrangement of Schwinger's proper time representation of the action. Nothing is changed. The new terms are calculated from the bound states arising from random potential wells generated by the path integral. Their calculation is analogous to finding the bound states of a charged fermion in a four-dimensional random magnetic field-four dimensional because the calculation is in Euclidean space. Such potentials are likely to contain broad, deep wells leading to closely spaced bound states. Every bound state contributes a negative term to the effective action, weighted by its degeneracy. The potentials generating these bound states are referred to as quasi-zero modes. Due to their probable large number, they can cause the effective action to turn negative from large amplitude-strong magnetic field-variations of the functional integral. The wells, being random, require no special symmetry. Such variations can cause the functional integral to converge.

Introduction. The question arises: can any QED process represented by a functional integral-path integral-be calculated non-perturbatively? As part of the electroweak model this question is of more than routine interest. The general belief is that it cannot due to the integral's presumed unbounded, large-field growth subsequent to QED's lack of asymptotic freedom. [An exception occurs when all fermion loops are neglected. But this is not QED.] Any attempt to elevate belief to certainty will require non-perturbative information on QED's effective action.

One has every reason to ask what can possibly be new here? After all, we are starting from work of Schwinger

dating from the 1950s. The answer is that a huge amount of information was overlooked in his 1951 paper [1] until years later. By simply rearranging Schwinger's expression for the effective action, new non-perturbative information was finally unlocked by mathematical theorems dating from the 1970s relating to constructive field theory [2–17]. The realization that some of them could be applied to four-dimensional QED came in 2015 [18]. It took eleven years to apply this information to the specific question above.

Consider, for example, the functional integral representation of the Euclidean Green function for $2n$ external fermions and m photons

$$S_{\mu_1, \dots, \mu_m}(x_1, \dots, x_n; y_1, \dots, y_n; z_1, \dots, z_m) = Z^{-1} \int d\mu_0(A) \det_{\text{ren}}(1 - eS\mathcal{A}) \det[S(x_i, y_j|eA)]_{i,j=1}^n \prod_{k=1}^m A_{\mu_k}(z_k), \quad (1)$$

where $S(x_i, x_j|A)$ is the fermion propagator in the external potential A_μ Z is the vacuum-to-vacuum amplitude that will eventually cancel out. The regularized fermion determinant $\det_{\text{ren}}$, written as $\exp(\ln \det_{\text{ren}})$, introduces the effective action $\ln \det_{\text{ren}}(1 - eS\mathcal{A})$, where S is the free fermion propagator. It is a non-linear, non-local gauge invariant quantity. Its expansion in powers of $eS\mathcal{A}$ generates the loop expansion which, when truncated, results in perturbation theory and is, ipso facto, unsuited to the above question. The subscript ren designates renormalization subtractions detailed below.

Further to [1], the potentials belong to a restricted class of random, tempered distributions belonging to $\mathcal{S}'(\mathbb{R}^4)$ that support the functional integral's Gaussian measure $d\mu_0(A)$. This requirement has been one of the main obstructions to the above question. Our aim is to review a representation of $\ln \det_{\text{ren}}$ [18] that can accommodate these potentials and provides non-perturbative estimates of its large-field variations. In this letter we show that its growth is slowed down due to the presence of quasi-zero modes embedded in the space of allowed potentials. Since we are calculating in Euclidean space we regard the field strength F to be static, four-dimensional magnetic field. A quasi-zero mode arises from a sequence of eigenvalues $\lambda_1, \lambda_2, \dots$ generated by solving for a random magnetic field's bound states. These are required to calculate the

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effective action. A sample magnetic field can generate a subset of eigenvalues, $\{\lambda_n\}$, that behave as $|\lambda_n| \rightarrow 1$ in the large-field limit. An eigenstate corresponding to one of these eigenvalues will be called QZM. Each of these modes reduce the growth of the effective action substantially.

The special case of vanishing binding energy is a zero mode that results in a negative contribution to the effective action without the need of a large-field limit. We expect these modes are sparse—requiring high symmetry—since the only known example of an admissible zero mode in the effective action of (1) requires an intricate $O(2) \times O(3)$ symmetry [18]. We will summarize known results relevant to the functional integral below.

In contrast, QZMs only require a deep, wide well in a random F . Randomness is the essential point here as there is apparently no limit to the number of QZMs. A random QZM's probable low symmetry–degeneracy–can be offset by the number of bound states supported by a random potential, thereby driving the effective action increasingly less positive for large field values. We now proceed to show this.

2 Potentials. These have already been discussed in [18] which dealt with the electrodynamic sector of the electroweak model. To alleviate any source of confusion here we outline the derivation of $\det_{\text{ren}}$ in QED alone to make this Letter reasonably self-contained. The potentials required to define $\det_{\text{ren}}$ and to support the functional measure $d\mu_0(A)$ in (1) are sufficient if (1) $A_\mu \in \mathcal{S}'(\mathbb{R}^4)$ with a $1/|x|$ falloff, (2) $A_\mu \in \bigcap_{r>4} L^r(\mathbb{R}^4)$, and (3) $F \in \bigcap_{r>2}^{<4} L^r(\mathbb{R}^4)$ assuming that the fermion mass $m \neq 0$ in $\det_{\text{ren}}$. The final requirement is to introduce an ultraviolet cutoff by smoothing the potentials $A_\mu \in \mathcal{S}'$ with the functions of rapid decrease f_Λ

$$A_\mu^\Lambda(x) = \int d^4y f_\Lambda(x-y) A_\mu(y), \quad (2)$$

so that $A_\mu^\Lambda \in C^\infty$ and hence is infinitely differentiable. A_μ^Λ and $F_{\mu\nu}^\Lambda = \partial_\mu A_\nu^\Lambda - \partial_\nu A_\mu^\Lambda$ also satisfy conditions (2)

and (3) above. The potentials A_μ have covariance

$$\int d\mu_0(A) A_\mu(x) A_\nu(y) = D_{\mu\nu}(x-y), \quad (3)$$

where $D_{\mu\nu}$ is the free photon propagator in a gauge fixed measure. Therefore the potentials A_μ^Λ have covariance

$$\int d\mu_0(A) A_\mu^\Lambda(x) A_\nu^\Lambda(y) = D_{\mu\nu}^\Lambda(x-y), \quad (4)$$

where, $D_{\mu\nu}^\Lambda$ has Fourier transform $\hat{D}_{\mu\nu} |\hat{f}_\Lambda(k)|^2$, with $\hat{f}_\Lambda(k) = 1$, $k^2 \leq \Lambda^2$, $\hat{f}_\Lambda(k) = 0$, $k^2 > \Lambda^2$. Thus the smoothing process also plays the dual role of an ultraviolet regulator Λ . In what follows all of the rough potentials A_μ in $\det_{\text{ren}}$ are replaced with the smooth potentials A_μ^Λ except in the measure $d\mu_0(A)$. From here on we will drop the superscript Λ ; it will only reappear again through its covariance (4), that is, when (1) is functionally integrated. We comment below on removing Λ .

We now state one of the most important results of this Letter, namely that the operator $\Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}} \in \mathcal{S}_3$, [2], [7], [18] where

$$\Delta_A^{\frac{1}{2}}(x, y) \equiv \frac{1}{\pi} \int_0^\infty \frac{da}{\sqrt{a}} \left\langle x \left| \left[(P - A)^2 + m^2 + a \right]^{-1} \right| y \right\rangle \quad (5)$$

and $\sigma_{\mu\nu} = [\gamma_\mu, \gamma_\nu]/2$, $\gamma_\mu^\dagger = -\gamma_\mu$. That this operator belongs to the trace ideal $\mathcal{S}_3$ means that it is compact with discrete eigenvalues having finite multiplicity. The integral (1) will give a non-perturbative result depending only on D^Λ since its measure is Gaussian. This should simplify the limit $\Lambda \rightarrow \infty$ of the result. Of course it is assumed that the integral (1) converges to which we now return.

3. $\det_{\text{ren}}$. We begin by recalling Schwinger's proper-time definition of $\ln \det_{\text{ren}}$ written as

$$\ln \det_{\text{ren}}(1 - eS\mathcal{A}) = \frac{1}{2} \int_0^\infty \frac{dt}{t} \left(\text{Tr} \left\{ e^{-P^2 t} - \exp \left[- \left(D^2 + \frac{e}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right\} + \frac{e^2 \|F\|^2}{24\pi^2} \right) e^{-tm^2}, \quad (6)$$

where $D_\mu = P_\mu - eA_\mu$, $\|F\|^2 = \int d^4x F^2$; e , m are unrenormalized; ultraviolet divergences are suppressed by the regulator Λ introduced above.

The last term in (6) is a charge renormalization subtraction in the one-particle irreducible photon self-energy at zero momentum transfer following Eq. (C7), Appendix C

[18]. It is extremely useful to rearrange (6) as

$$\begin{aligned} \ln \det_{\text{ren}} = & \frac{1}{2} \int_0^\infty \frac{dt}{t} e^{-tm^2} \left[4 \text{Tr} \left(e^{-P^2 t} - e^{-D^2 t} \right) - \frac{e^2 \|F\|^2}{48\pi^2} \right. \\ & \left. + \text{Tr} \left(e^{-D^2 t} - \exp \left[- \left(D^2 + \frac{e}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right) + \frac{e^2 \|F\|^2}{16\pi^2} \right], \end{aligned} \quad (7)$$

where the trace over spin was made in the first term to give a factor of 4. A volume cutoff on $\|F\|^2$ will be

discussed below. Then (7) becomes

$$\begin{aligned} \ln \det_{\text{ren}} &= 2 \ln \det_{\text{SQED}} + \frac{1}{2} \int_0^\infty \frac{dt}{t} e^{-tm^2} \\ &\times \left[\text{Tr} \left(e^{-D^2 t} - \exp \left[- \left(D^2 + \frac{e}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right) \right. \\ &\left. + \frac{e^2 \|F\|^2}{16\pi^2} \right]. \end{aligned} \quad (8)$$

Here $\ln \det_{\text{SQED}}$ is the proper-time definition of the formal scalar QED determinant $\ln \left\{ \left[(P-A)^2 + m^2 \right] / (P^2 + m^2) \right\}$ with on-shell charge renormalization

$$\ln \det_{\text{SQED}} = \int_0^\infty \frac{dt}{t} \left[\text{Tr} \left(e^{-P^2 t} - e^{-D^2 t} \right) - \frac{e^2 \|F\|^2}{192\pi^2} \right] \times e^{-tm^2}. \quad (9)$$

Care is needed to make sense of the remaining trace in (8) subject to the requirements (2) and (3) on the potentials listed above. This is done by breaking it up into two well-defined terms. Introduce the operator identity [19]

$$e^{-t(D^2 + \frac{1}{2}e\sigma F)} - e^{-tD^2} = \int_0^t ds e^{-(t-s)(D^2 + \frac{1}{2}e\sigma F)} \frac{1}{2} e\sigma F e^{-sD^2}, \quad (10)$$

and iterate it twice to get terms of order σF and $(\sigma F)^2$ plus the remainder term. Insert these in (8). The σF term in the expansion cancels in (8) due to the spin trace. The remainder term in (8) defines the determinant $\det_3$ as

$$\begin{aligned} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e\sigma F \Delta_A^{\frac{1}{2}} \right) &= \\ \int_0^\infty \frac{dt}{t} e^{-tm^2} \text{Tr} &\left[\int_0^t ds_1 \int_0^{t-s_1} ds_2 \int_0^{t-s_1-s_2} ds_3 \right. \\ &\times e^{-(t-s_1-s_2-s_3)(D^2 + \frac{1}{2}e\sigma F)} \\ &\times \frac{1}{2} e\sigma F e^{-s_3 D^2} \frac{1}{2} e\sigma F e^{-s_2 D^2} \frac{1}{2} e\sigma F e^{-s_1 D^2} \left. \right], \end{aligned} \quad (11)$$

with $\Delta_A^{\frac{1}{2}}$ defined in (5). This introduces the regularized determinant

$$\det_3 (1 + T) = \det \left[(1 + T) \exp \left(-T + \frac{1}{2} T^2 \right) \right], \quad (12)$$

where $T \in \mathcal{S}_3$. Its trace converges and depends on T^3 and higher powers of T . The equivalence of the two sides of (11) follows from Theorem 7.2 in [6]. Because of its inaccessibility and printing errors we rederive it in Appendix B of [18]. Combining (9), (11) and the second iterate of (10) results in

$$\begin{aligned} \ln \det_{\text{ren}} &= 2 \ln \det_{\text{SQED}} + \frac{1}{2} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e\sigma F \Delta_A^{\frac{1}{2}} \right) \\ &+ \frac{e^2}{8} \int_0^\infty \frac{dt}{t} e^{-tm^2} \left(\frac{1}{4\pi^2} \|F\|^2 - \text{Tr} \int_0^t ds_1 \int_0^{t-s_1} ds_2 e^{-(t-s_1-s_2)D^2} \sigma F e^{-s_2 D^2} \sigma F e^{-s_1 D^2} \right). \end{aligned} \quad (13)$$

The last term simplifies following Appendix C of [18] to give the three-term representation of $\det_{\text{ren}}$.

$$\begin{aligned} \ln \det_{\text{ren}} &= 2 \ln \det_{\text{SQED}} + \frac{1}{2} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e\sigma F \Delta_A^{\frac{1}{2}} \right) \\ &+ e^2 \int_0^\infty dt e^{-tm^2} \\ &\times \left[\frac{1}{32\pi^2 t} \|F\|^2 - \frac{1}{2} \text{Tr} \left(e^{-tD^2} F_{\mu\nu} \Delta_A F_{\mu\nu} \right) \right]. \end{aligned} \quad (14)$$

It is equivalent to (6) following its rearrangement. Since the operator in $\det_3$ belongs to $\mathcal{S}_3$ we can use the operator identity

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$$\begin{aligned} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e\sigma F \Delta_A^{\frac{1}{2}} \right) &= \ln \det \left[(1 + T) \exp \left(-T + \frac{1}{2} T^2 \right) \right] \\ &= \text{Tr} \left[\ln (1 + T) - T + \frac{1}{2} T^2 \right] \\ &= \sum_{n=1}^\infty \left[\ln (1 - \lambda_n) + \lambda_n + \frac{1}{2} \lambda_n^2 \right]. \end{aligned} \quad (15)$$

The eigenvalues $\{\lambda_n\}$ of $\Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}}$ occur in pairs $(-\lambda, \lambda)$ following C -invariance (Appendix D, [18]). We can alternatively replace $\ln \det_3$ in [14] with an eigenvalue sum

$$\ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e\sigma F \Delta_A^{\frac{1}{2}} \right) = \sum_{n=1}^\infty [\ln (1 - \lambda_n^2) + \lambda_n^2], \quad (16)$$

where the sum is only over positive eigenvalues. Since $\ln \det_3$ is real and finite then $0 < \lambda_n < 1$ for all n so

that $\ln \det_3 < 0$. The sum in (16) has an infinite radius of convergence. That $\ln \det_3 < 0$ is due to the paramagnetism of a charged spin- $\frac{1}{2}$ fermion in a static, four dimensional magnetic field. On average the eigenvalues are lowered by $F_{\mu\nu}$ following the Peierls-Bogoliubov inequality derived from Klein's inequality [20–22]. It is already apparent in (15) and (16) that the lowest-lying eigenvalues will dominate the convergence of (1) in the large magnetic field limit due to QZMs defined above. That these were overlooked since 1951 is probably because they are essentially non-perturbative.

4. Analytic calculation of a QZM. The eigenvalues of (16) are obtained from

$$\frac{e}{2} \Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}} \varphi_n = -\lambda_n \varphi_n, \quad (17)$$

where $\varphi_n \in L^2$. Let $\Delta_A^{\frac{1}{2}} \varphi_n = \psi_n$ and obtain

$$\left[(P - eA)^2 + \frac{e}{2\lambda_n} \sigma F \right] \psi_n = -m^2 \psi_n, \quad (18)$$

where $\psi_n \in L^2$ as shown at the end of Appendix A of [18]. From here on we assume e is real. The $\{\lambda_n\}$ assume the role of coupling constants whose discrete values result in bound states with energy $-m^2$. The states ψ_n have definite chirality since γ_5 commutes with σ by choosing a representation of γ_5 with diagonal elements $\pm \mathbb{1}_2$. Therefore we need only deal with 2D chiral eigenstates $\psi_n^\pm$. To demonstrate an analytically solvable QZM we choose an $O(2) \times O(3)$ symmetric potential. These have the standard form [23–25].

$$A_\mu(x) = M_{\mu\nu} x_\nu a(r), \quad (19)$$

here $M_{\mu\nu}$ is the antiself-dual, antisymmetric matrix with nonvanishing elements $M_{12} = M_{30} = 1$ and $r^2 = x_\mu^2$. This potential is chosen solely to simplify the analysis; QZMs have no particular symmetry. The eigenstates of (18) are represented by a 1×4 column matrix [25] whose first two rows consist of the positive chirality states

$$\psi_n = r^{-2j-\frac{3}{2}} \begin{bmatrix} \mathcal{D}_{M-\frac{1}{2},m}^j \rho_1(r) \\ \mathcal{D}_{M+\frac{1}{2},m}^j(x) \rho_2(r) \end{bmatrix}, \quad (20)$$

where $\mathcal{D}_{m_1, m_2}^j$ are the four dimensional rotation matrices [25–27] and $2j = 0, 1, \dots; -j \leq m \leq j$. Inserting (20) into (18) gives

$$\left[-\frac{d^2}{dr^2} + \frac{(2j+1)^2 - \frac{1}{4}}{r^2} + (4M \mp 2)ea + e^2 r^2 a^2 \right. \\ \left. \pm \frac{e}{\lambda_n^\pm} \left(4a + r \frac{da}{dr} \right) \right] \rho_{1,2} = -m^2 \rho_{1,2}, \quad (21)$$

where the upper (lower) sign applies to ρ_1 (ρ_2) and $\lambda_n^\pm$ denotes a positive chirality eigenvalue that results in L^2 bound states. In fact it is already evident from (18) that

only $0 < \lambda_n < 1$ results in a bound state when e is real. We chose one of the simplest potentials (19) with

$$a(r) = \begin{cases} \frac{C}{R^2} \left(\frac{r}{R} \right)^{\varepsilon-2} + \frac{(2-\varepsilon)C-2\nu}{R^3} r + \frac{(\varepsilon-3)C+3\nu}{R^2} & \text{if } r \leq R \\ \frac{\nu}{r^2}, & r > R \end{cases} \quad (22)$$

giving $F_{\mu\nu}^2 = 8r^2 a'^2 + 16a^2 + 16r^2 a a'$, $a' = \frac{da}{dr}$, and $\varepsilon \geq 2$ so that $F \in \bigcap_{p>2} L^p$. The constant C can be positive or

negative, and $\nu > 0$. In order that $A_\mu \in C^\infty$ we assume that there is an infinitesimal zone of smooth curvature centered at $r = R$. This has no effect on the present analysis. Setting $M = -j - \frac{1}{2}$ to ensure a bound state in (21) and letting $\rho_2 = f$ with $r \leq R$, $e \gg 1$.

$$\left[\frac{d^2}{dr^2} + \frac{\frac{1}{4} - (2j+1)^2}{r^2} + 4 \left(j + \frac{1}{\lambda} \right) ea \right. \\ \left. - e^2 r^2 a^2 + \frac{e}{\lambda} r \frac{da}{dr} \right] = m^2 f. \quad (23)$$

For $r > R$ let $f = r^{\frac{1}{2}} g$ so that (23) becomes

$$g'' + \frac{1}{r} g' - \left(m^2 + \frac{(2j+1-\varepsilon\nu)^2 + 2(1-\frac{1}{\lambda})\varepsilon\nu}{r^2} \right) g = 0. \quad (24)$$

This has the decaying solution $K_\alpha(mr)$ which is a modified Bessel function with $\alpha = \left[(2j+1-\varepsilon\nu)^2 + 2(1-\frac{1}{\lambda})\varepsilon\nu \right]^{\frac{1}{2}}$. The eigenvalue λ is fixed by the boundary condition at $r = R$. The analysis is simplified by setting $mR \ll 1$ and making the small mass expansions $f = B f_0 (1 + m^2 f_2 + \mathcal{O}(m^4 f_4))$, $\lambda = 1 - m^2 \delta_2 + \mathcal{O}(m^4 \delta_4)$, B is a normalization constant. Here f_0 is the solution of (23) when $m = 0$ and $\lambda = 1$ and $0 \leq r \leq R$ where $f_0 = r^{2j+\frac{3}{2}} e^{-e \int_0^r ds a(s)}$. The two sides of the joining condition can be matched in powers of m to obtain for $\varepsilon\nu \gg 1$

$$\lambda = 1 - \frac{m^2 R^2}{2\varepsilon\nu} \left(1 + \mathcal{O}\left(\frac{1}{\varepsilon\nu} \right) \right), \quad (25)$$

following Appendix E of [18]. Such an eigenvalue approaching 1 from below is the signature of a QZM as referred to in the introduction above.

We conclude that for this mode its contribution to $\ln \det_3$ in (16) is

$$\ln \det_3 = -\frac{e^2 \nu^2}{2} \ln \left(\frac{\varepsilon\nu}{m^2 R^2} \right) \left(1 + \mathcal{O}\left(\frac{1}{\varepsilon\nu} \right) \right) \\ + \sum_{n=2}^{\infty} [\ln(1 - \lambda_n^2) + \lambda_n^2]. \quad (26)$$

All remaining bound states from (19), (22) contribute a negative term to (26). Presumably (25) is the lowest lying bound state eigenvalue. There might remain a band of closely spaced bound states for $n \geq 2$, each with degeneracy of $\mathcal{O}(e^2 \nu^2)$. All of the bound states of (17)

contribute a negative term to $\ln \det_3$ irrespective of their symmetry or lack thereof.

5. Completion of $\det_{\text{ren}}$. The first and third terms in (14) have already been calculated in the large amplitude limit in (1). From Eq. (A24) in [24] we found

$$\ln \det_{\text{SQED}} \stackrel{e\mathcal{F} \rightarrow \infty}{\sim} -\frac{e^2 \|F\|^2}{192\pi^2} \ln \left(\frac{e\mathcal{F}}{m^2} \right) + \mathcal{O}(e\mathcal{F}), \quad (27)$$

where the amplitude of $F_{\mu\nu}$ is represented by the parameter $\mathcal{F}$ with dimension L^{-2} . The third term is referred

to here as the charge renormalization term, denoted by Π in Eq.(6.43) in [18] yielding

$$\Pi \stackrel{e\mathcal{F} \rightarrow \infty}{\sim} -\frac{e^2 \|F\|^2}{32\pi^2} \ln \left(\frac{e\mathcal{F}}{m^2} \right) + R_1, \quad (28)$$

with $\lim_{e\mathcal{F} \rightarrow \infty} R_1 / [(e\mathcal{F})^2 \ln(e\mathcal{F})] = 0$. In order to compare (27) and (28) with $\ln \det_3$ in (21) note that $\max F_{\mu\nu}^2 = \frac{8\nu^2}{R^4} \equiv \mathcal{F}^2$ for $0 \leq r < \infty$. Accordingly, we set $e\mathcal{F} = \frac{e\nu}{R^2}$ for $e\nu \gg 1$. Then from (14), (26), (27) and (28)

$$\begin{aligned} \ln \det_{\text{ren}} \stackrel{e\mathcal{F} \rightarrow \infty}{\sim} & \frac{e^2 \|F\|^2}{32\pi^2} \ln \left(\frac{e\mathcal{F}}{m^2} \right) + R_1 - \frac{e^2 \|F\|^2}{196\pi^2} \ln \left(\frac{e\mathcal{F}}{m^2} \right) + \mathcal{O}(e\mathcal{F}) \\ & - \frac{e^2 \mathcal{F}^2 R^4}{16} \ln \left(\frac{e\mathcal{F}}{\sqrt{8}m^2} \right) \left[1 + \mathcal{O} \left(\frac{1}{e\nu} \right) \right] + \sum_{n=2}^{\infty} [\ln(1 - \lambda_n^2) + \lambda_n^2]. \end{aligned} \quad (29)$$

Additional zero mode contributions to $\ln \det_{\text{ren}}$ such as (6.45), (6.46) in [18] are omitted as they are not random variables and therefore not relevant here.

6. $\det_3$: General properties. (a) $\det_3(1 + eT)$, $T = \Delta_{\frac{1}{2}A}^{\frac{1}{2}} \sigma F \Delta_{\frac{1}{2}A}^{\frac{1}{2}}$ is well defined by a series for $|e|$ small and defined by analytic continuation to an entire function of e considered as a complex variable, which means that it is holomorphic in the entire complex plane [4]; (b) Let $\lambda_1(T), \dots$ be the eigenvalues of $T \in \mathcal{S}_3$. Then $\sum_{n=1}^{\infty} |\lambda_n|^{-3} < \infty$ [4]. The reality of $\det_3$ for real e together with C -invariance require that the eigenvalues appear as real pairs $\lambda_n, -\lambda_n$ or as quartets $\pm\lambda_n, \pm\bar{\lambda}_n$ for $m \neq 0$; (c) $\det_3$ removes the $\mathcal{O}(e)$ term. The remaining $\mathcal{O}(e^2)$ term is replaced with the on-shell renormalized term Π whose asymptotic result is (6.43) in [18]. This result was combined with $\ln \det_{\text{SQED}}$ in (27); (d) The growth of $\det_3$ for $|e|\mathcal{F} \rightarrow \infty$ depends on the distribution of its zeros $-\frac{1}{\lambda_1}, \dots$ [28].

7. Conclusion. The opening question of this Letter remains unanswered. Progress has been made here by narrowing it to Eq. (29). Its first two terms sum to $+e^2 \|F\|^2 / 48\pi^2 \ln \left(\frac{e\mathcal{F}}{m^2} \right)$, which is Weisskopf's 1936 strong-field result when continued to Euclidean space [29]. Loosely stated, this limit $e\mathcal{F} \gg m^2$ of (29) gives

$$\ln \det_{\text{ren}} \sim \ln \det [\text{Weisskopf, 1936}] + \ln \det_3. \quad (30)$$

The last term, $\ln \det_3$, is always negative, and hence is the controlling determinant for asymptotic freedom in QED. The $\ln(e\mathcal{F}/m^2)$ factor in $\det_3$ is always present due to QZMs. Furthermore, it does not require a volume cutoff. We know that the distribution of the eigenvalues of the compact operator $T \in \mathcal{S}_3$ in the complex e -plane determines the growth of $|\det_3|$ for $|e\mathcal{F}| \rightarrow \infty$ [28]. Knowing this should be enough to settle the question of asymptotic freedom in QED.

A convergent result for a QED process (1) will contain an ultraviolet cutoff Λ in the free photon propagator following the Gaussian integration (3). Recall that it was introduced to smooth the tempered distributions supporting the functional integral. A volume cutoff also remains. The ability to remove these cutoffs by some non-perturbative renormalization procedure leads directly to the long-standing question of whether QED is an interacting theory.

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